

Case studies on smart classes

According to a report by Inderpreet Kaur Chachra on the effectiveness of smart learning, published in June 2015 segment of Abhinav National Monthly Refereed Journal of Research in Arts & Education, stats shown on the right were observed.

This performance improvement is more significant in the above- and below-average students. See below for details.

Category	Method of teaching	Number	Performance	SD
Below-average students	Smart classroom	10	6.7	.95
	Traditional teaching	10	4.8	.79
Above-average students	Smart classroom	10	9.6	.52
	Traditional teaching	10	8.1	.99

Studies conducted at numerous other schools and colleges provide similar results. With the addition of smart learning technologies, an improvement of student performance is the general trend.

on home automation and students can easily make home automation products for earning good money," added Jain.

RIOT is one of the popular free operating systems for the development of IoT projects. Tools such as these are developed by a grassroots community, thus helping reduce product development costs. People have started using technology in fields that usually do not care for it, because of the significantly-lowered (it does not get lower than free) cost of entry.

Similarly-named RIOT is an ARM based board that is available for IoT developers. Variants of Arduino boards continue to be very popular for IoT projects, too.

Pathy Iyer, head of business development at Keysight Technologies, says, "Speaking about the IoT, we are already setting up a lab in NASSCOMM where students can come over and try out their designs. Device-to-device communications and new protocols specific to the IoT that are emerging can be tested here."

Whatever happened to the college lab

Any electronics lab in a college would have at least a simple, low-frequency function generator for performing the basic electronics experiments. Similar basic test and measurement tools continue to sell, as usual. While conventional labs continue to exist in a majority of colleges, we are now seeing an increased interest in the setting up of remote labs and virtual labs in colleges.

"Equipment-wise, one area where we are seeing good interest coming from colleges is on the devices and semiconductor side.

The new semiconductor fab coming up in Gujarat is a good example of something that can drive jobs in this high-technology sector. We are partnering with Hindustan Semiconductor Manufacturing (HSMC) for this," adds Iyer. Arbitrary waveform generation, oscilloscopes, RF signal and function generators, design tools like Advanced Design System (ADS) and SystemVue electronic system-level design are what Iyer sees to be the most required.

3D printers have also made a strong presence in college labs, led by forward-thinking academia. B.M. Shah, proprietor, Edkits Electronics, says, "Costing for 3D printers has gone low, hence it is impacting big time."

Prof. Arun Giriapur, head - automation and robotics, BVB College of Engineering and Technology, adds, "Students can now create the final product. They can measure length of wires, position of components and heat dissipation to create optimal products via the 3D printer."

This allows for T-shaped skill development in the students. Giriapur explains, "We have teams consisting students from various streams collaborating on projects to increase their exposure to different engineering segments."

There is an interest towards mechatronics in students for graduation. It combines traits from most engineering streams to develop a new species of engineers. "Students are developing interest in mechatronics on a scale larger than ever before," says Compella.

Shah adds, "Manufacturing plant monitoring using a Wi-Fi camera along with a development board can help in remote applications." This move to mechatronics has been in large due to the various equipment



An electronic work-bench